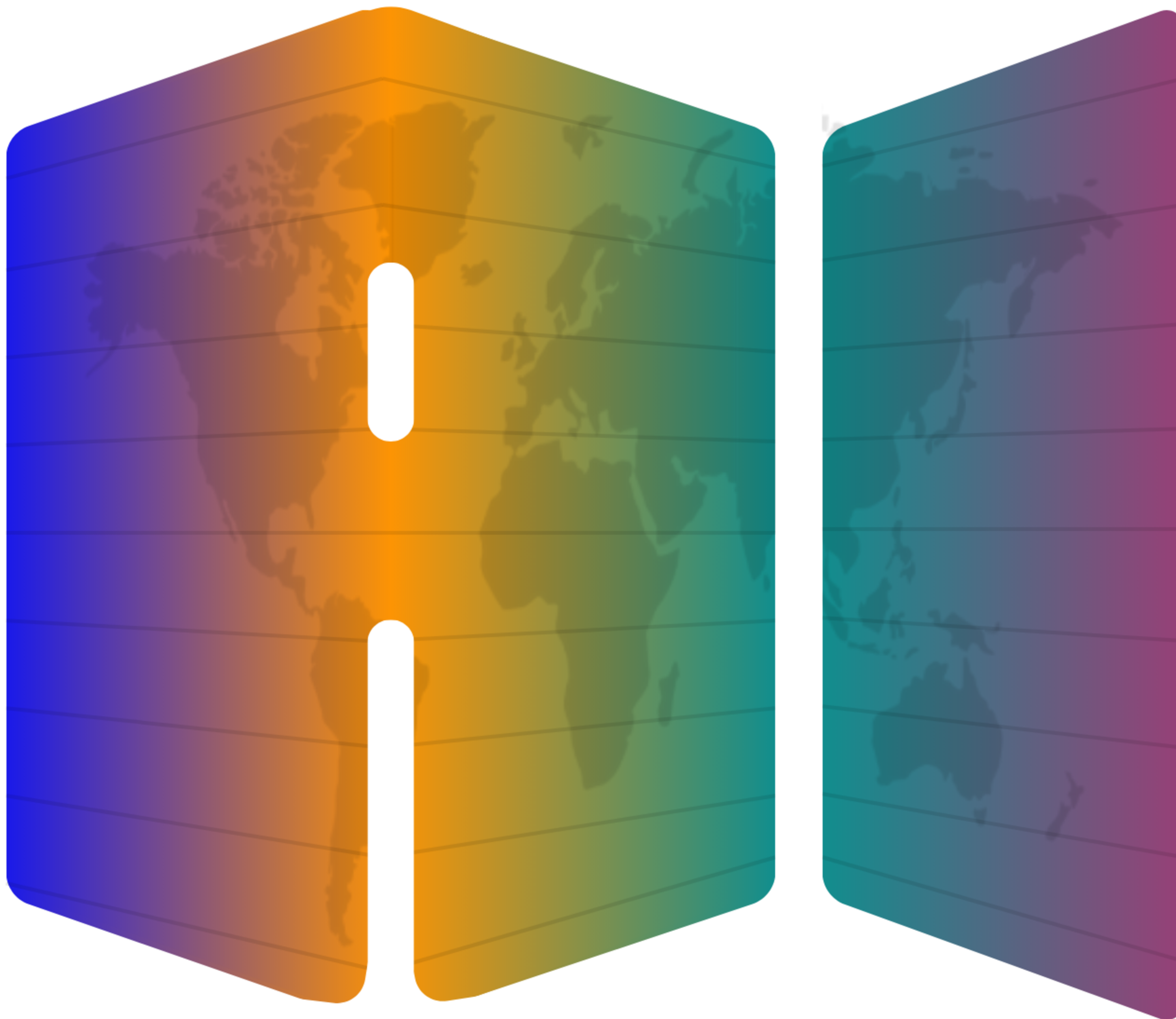


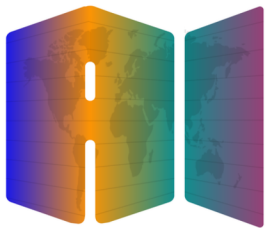


Agent2Agent Protocol

Lectures by Holt Skinner

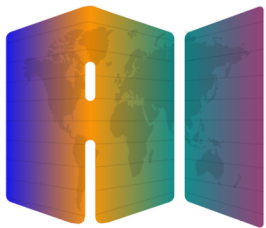
Ivan Nardini

Sandi Besen

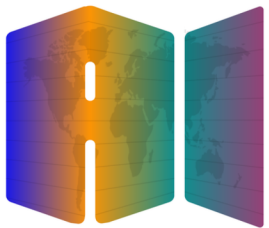
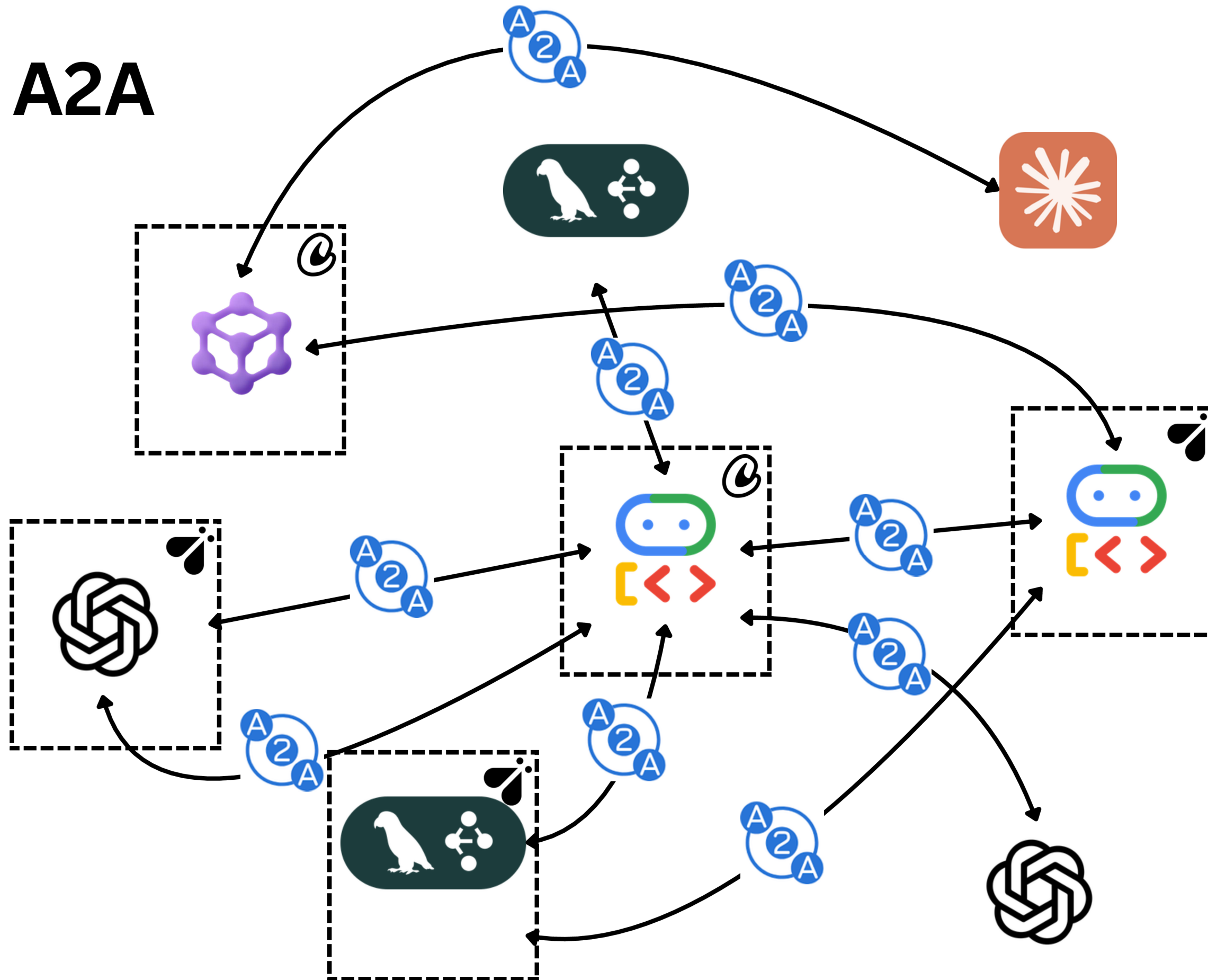


Agent2Agent Protocol

Why A2A Protocol?

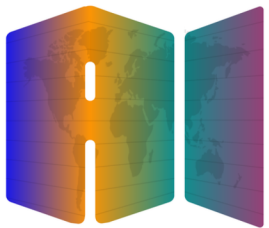


Why A2A



Why A2A

It provides a standardized approach
for agents to **interact**



Why A2A

A2A Capabilities



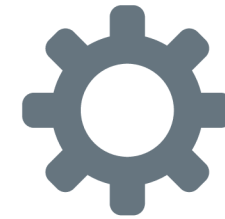
Discovery

Agents must advertise their capabilities so clients know when and how to utilize them for specific tasks



Negotiation

Clients and agents need to agree on communication methods like text, forms, iframe, or audio/video to ensure proper user interaction



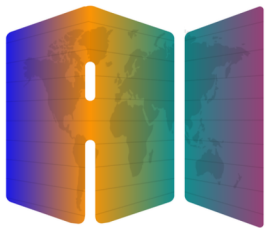
Task and State Management

Clients and agents need mechanisms to communicate task status, changes, and dependencies throughout task execution



Collaboration

Clients and agents must support dynamic interaction, enabling agents to request clarifications, information, or sub-actions from client, other agents, or users



Why A2A

A2A ❤️ MCP Complementary Protocols



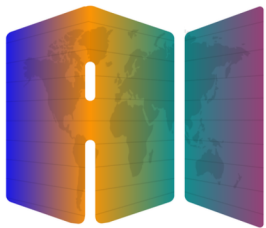
Model Context Protocol

- Connects agents to tools, APIs, and resources
- Think: *How an agent uses its capabilities (function calling)*
- Example: Agent uses MCP to call a weather API tool.

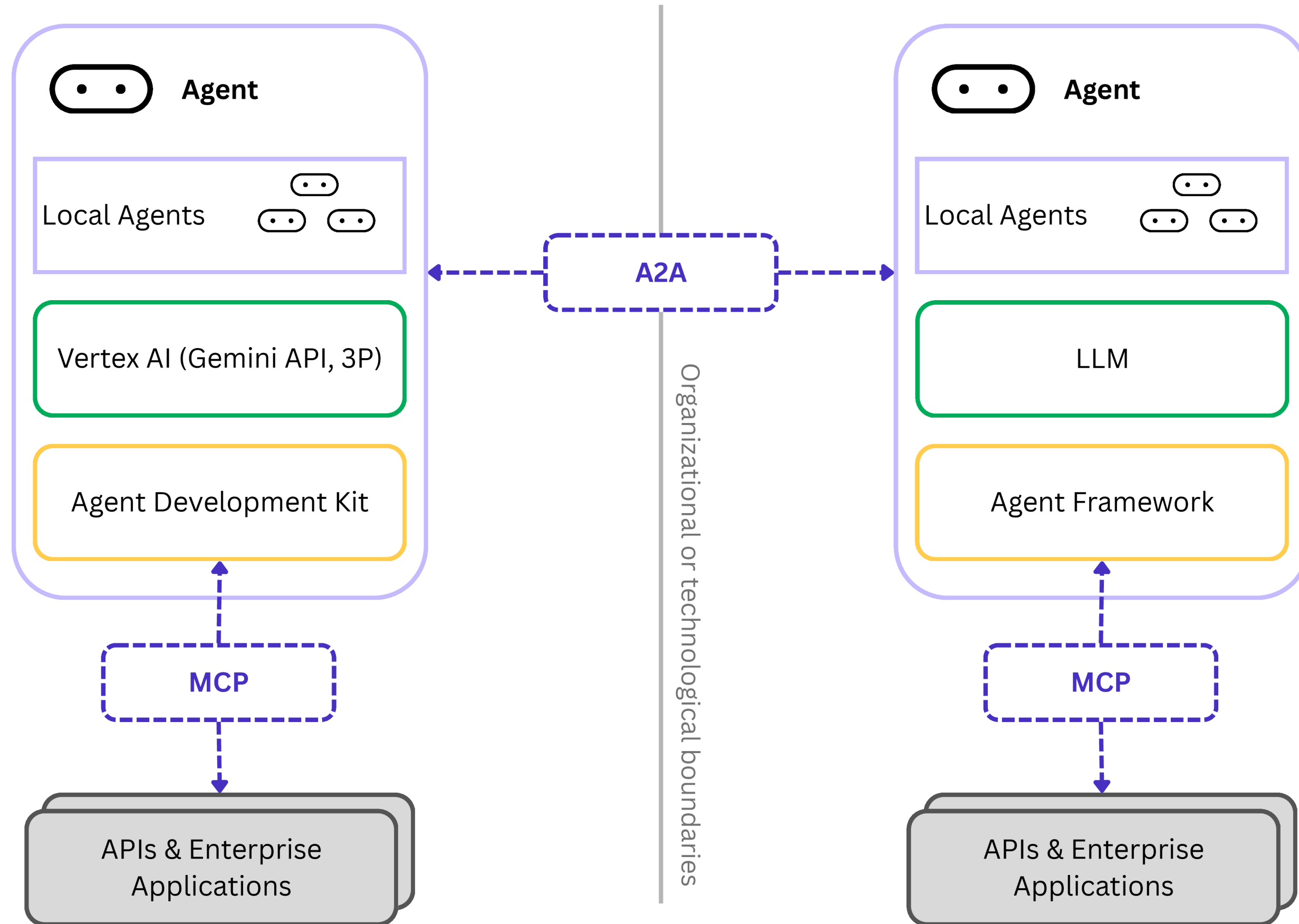
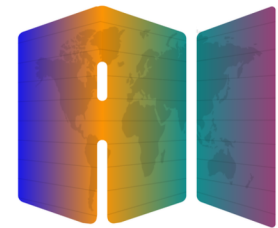


Agent2Agent Protocol

- Facilitates dynamic communication between different agents as peers.
- Think: *How agents collaborate, delegate, and manage shared tasks.*
- Example: A Travel Agent asks a Flight Booking Agent to find flights



Why A2A



Why A2A

Agents or Tools



Philip_Stephens Google Staff

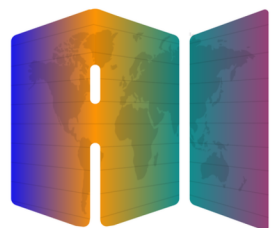
2 ✎ Jun 2025

There is a common refrain from many that “why can’t agents just be tools?”. This blog provides a point of view about why tools and agents should be treated differently, and why there needs to be a different way to interact with them.

When asking about whether agent interoperation is simply just using an agent as a tool in another agent, we need to define the types of operational boundaries that exist. To do this we must move our thinking to the source of the control flow decisions and the scope that decisions are to be made in.

The well-defined nature of a tool

Consider the case of an agent that can utilize multiple tools. The tools are used to take action, either gather information, make a change or to transform information from one form to another. In each of these cases, there is a temporal relationship that input is provided and some output is returned. This is illustrated in the figure below, where there is also an example of streaming action.



goo.gle/agents-not-tools

Why A2A

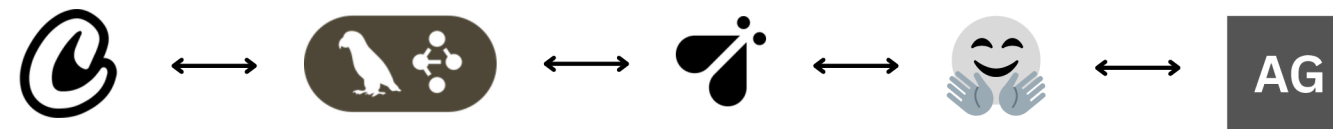
Where A2A Fits in the Tech Stack

Applications



Agentic orchestration

Agent2Agent (A2A) Protocol



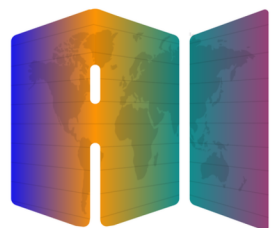
Foundation models



Cloud Infrastructure

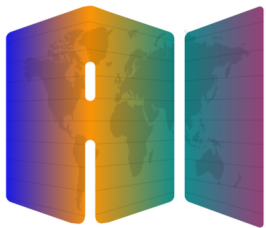


Semiconductors



Agent2Agent Protocol

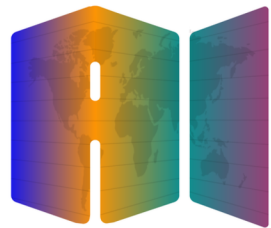
A2A Architecture



A2A Architecture

Common Language for Agents

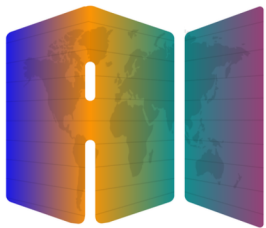
A2A utilizes standard web technologies like HTTP, JSON-RPC, gRPC, and Server-Sent Events (SSE) to communicate between independent systems



A2A Architecture

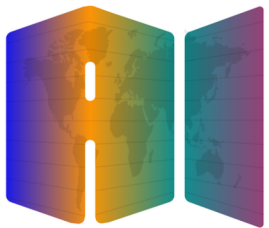
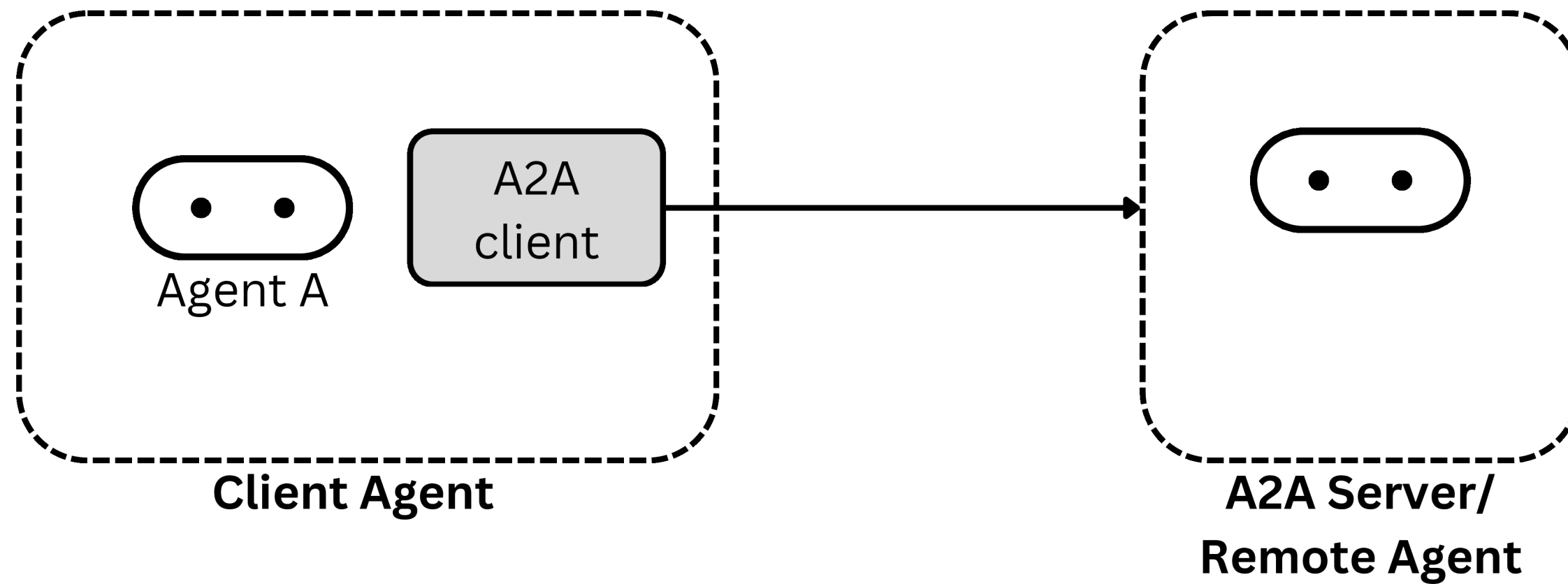
Common Language for Agents

a2a-protocol.org/latest/



A2A Architecture

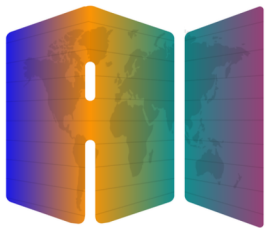
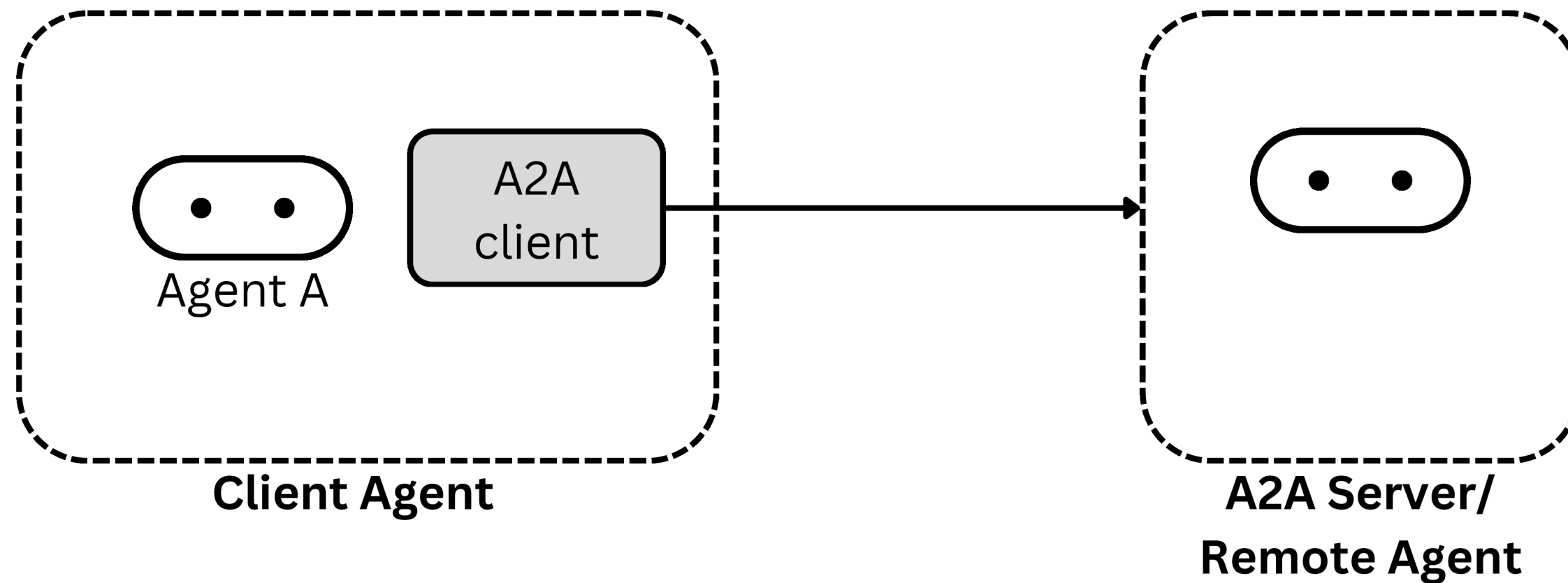
Building a simple agent system



A2A Architecture

Building a simple agent system

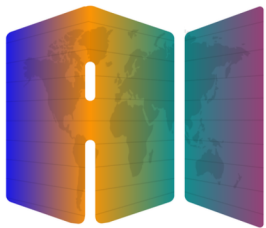
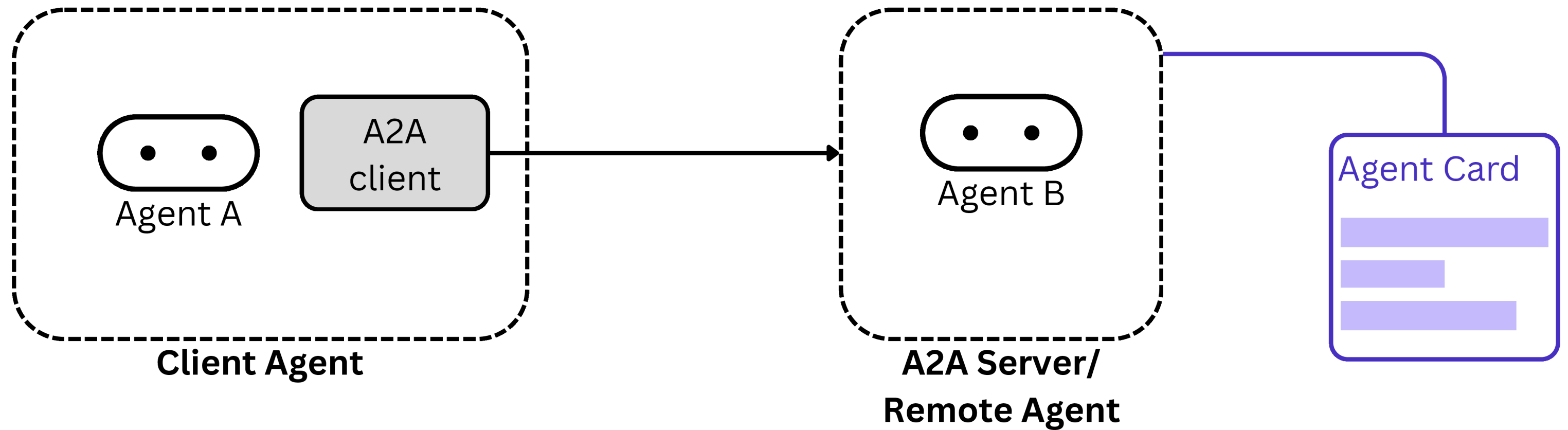
Who are you and what can you do?



A2A Architecture

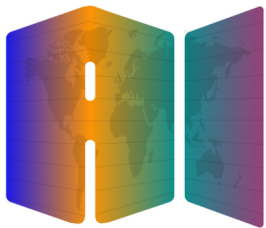
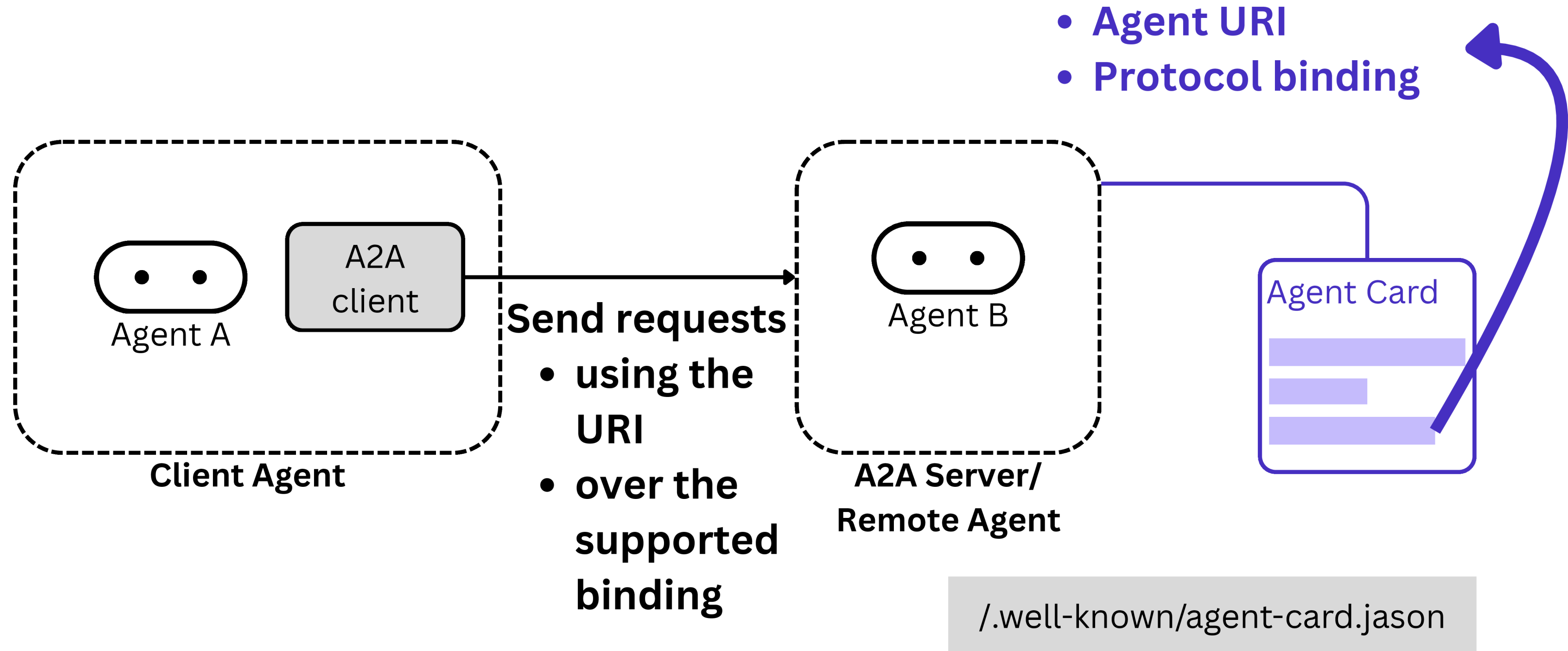
Agent Discovery

The Agent Card



A2A Architecture

How do they actually talk?



A2A Architecture

Execution Modes

Synchronous

Waiting for an immediate response

Asynchronous

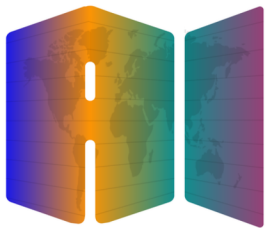
Proceeding without blocking for a reply

Streaming

Transmitting continuous data flows

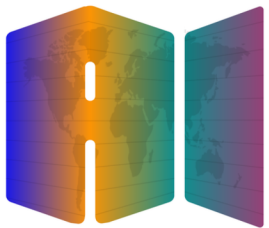
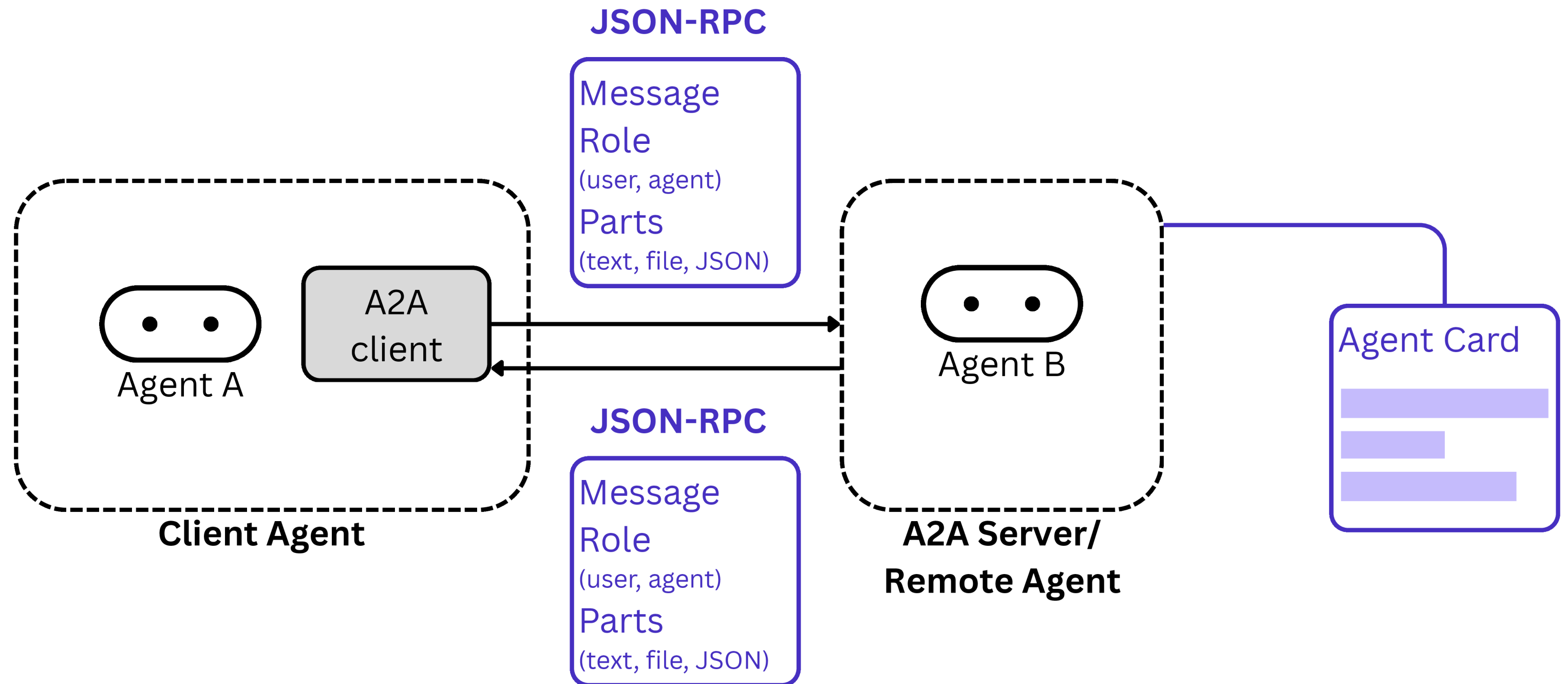
Push Notifications

Alerting the other agent when specific events occur



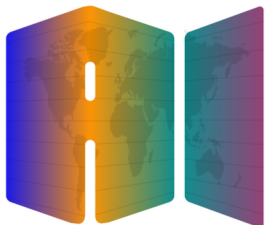
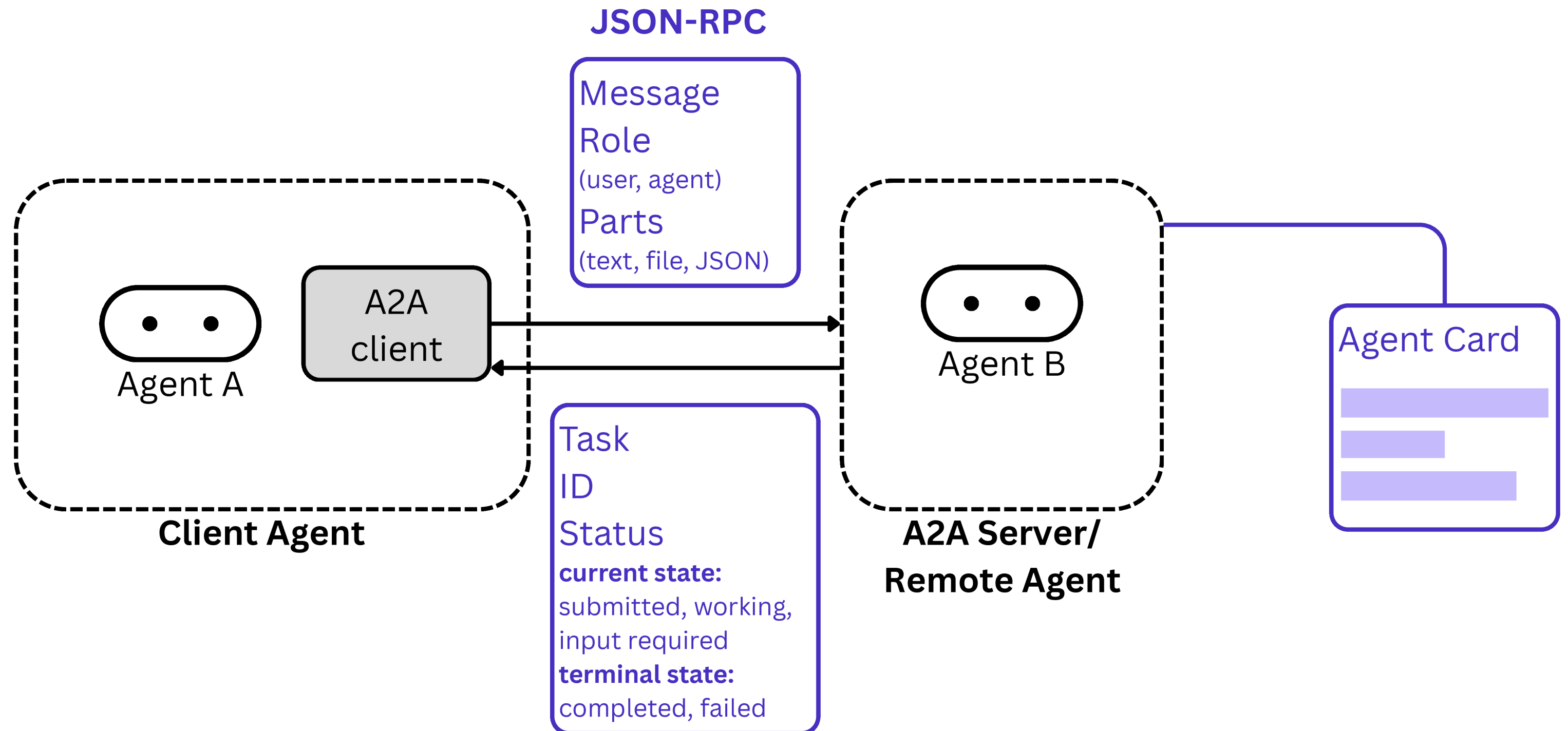
A2A Architecture

Synchronous Interaction



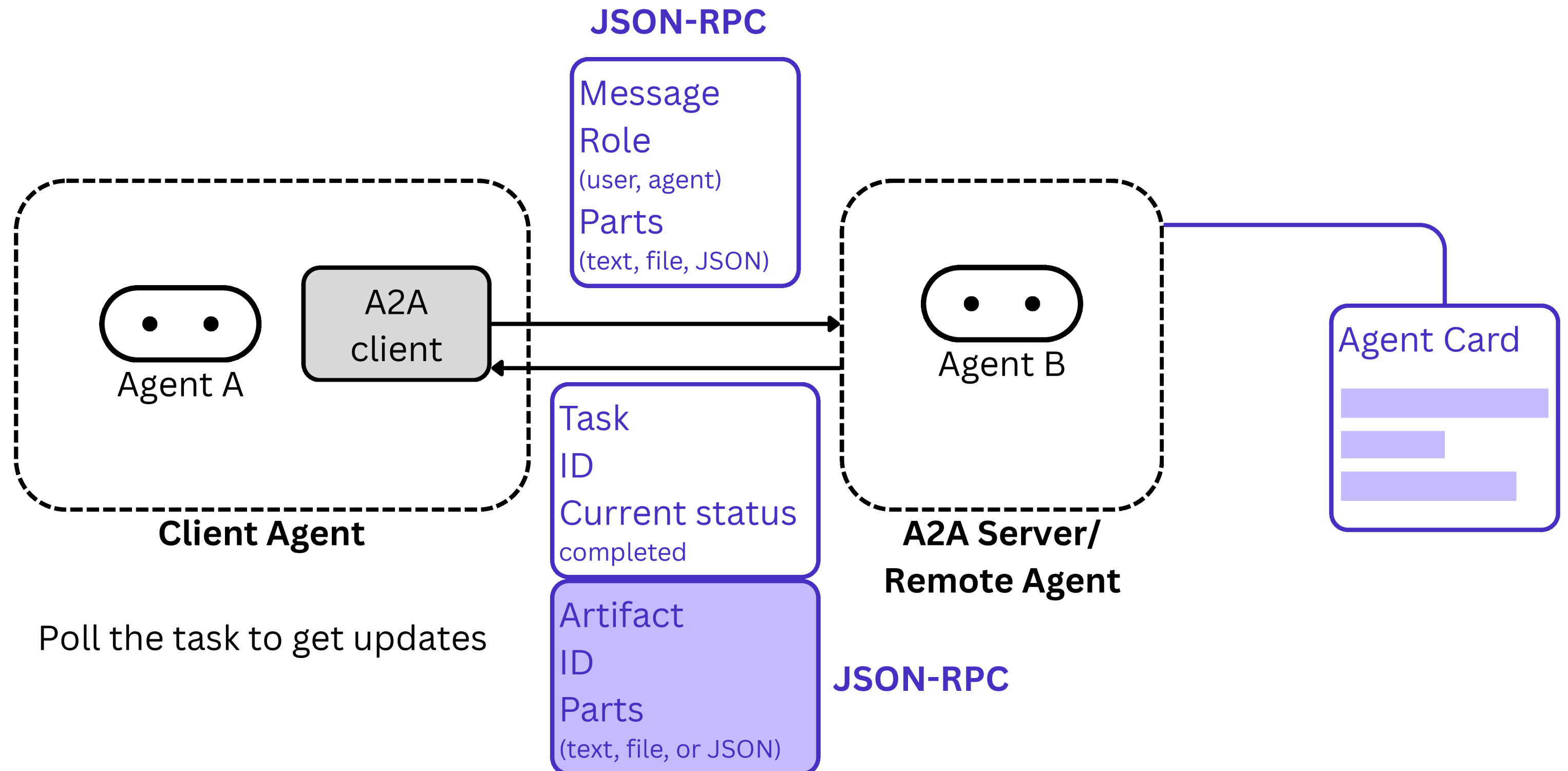
A2A Architecture

Asynchronous Communication



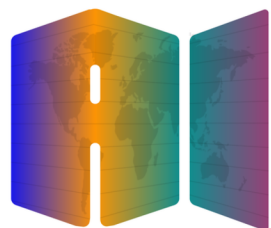
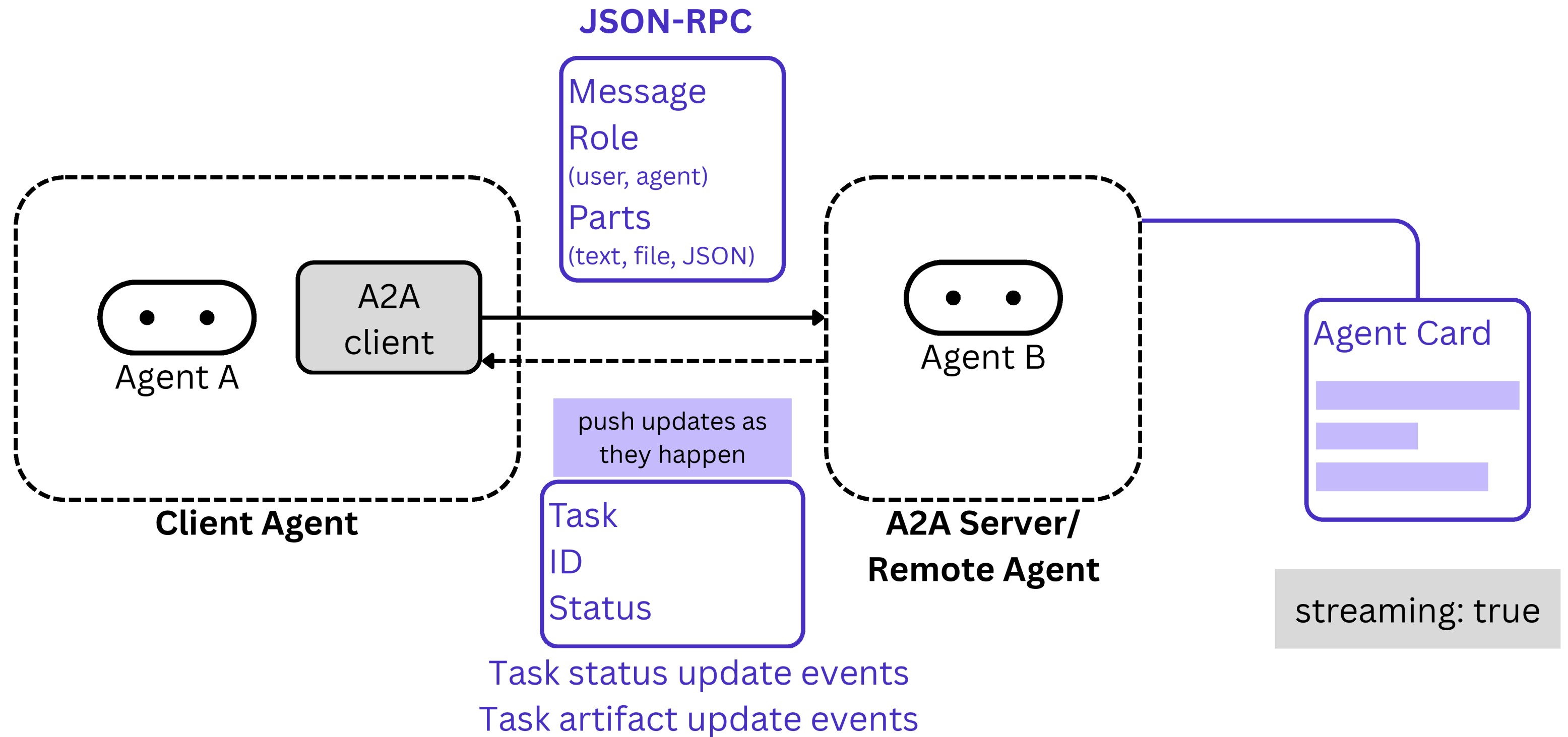
A2A Architecture

Asynchronous Communication



A2A Architecture

Streaming with Server-Sent Events (SSE)



A2A Architecture

Push Notifications

Webhooks & Callbacks

